LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Classification of aircraft parts, types of aircraft fuselage structure, joining techniques, riveted joints, bonded joints, brazing.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Classification of aircraft parts, types of aircraft fuselage structure, joining techniques, riveted joints, bonded joints, brazing.
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of joint types on L & Ds; Safety precautions regarding heat and tools; Instructor supervision during model analysis.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary structural elements of an aircraft?
(b) List three common methods used to join aircraft parts.
(c) Define the term 'fuselage' in the context of aircraft structure.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Classification of aircraft parts • Types of fuselage structure • Riveted joints types • Bonded joints characteristics • Brazing techniques	L & Ds	PPT, IB	15 min
(b)	• Monocoque vs Semi-monocoque • Spars and ribs function • Rivet types and spacing • Adhesive bonding process • Brazing filler metals	L & Ds	PPT, PDF	20 min
(c)	• Joint efficiency factors • Inspection of bonded joints • Brazing temperature control • Corrosion prevention methods • Maintenance of structural integrity	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a monocoque and semi-monocoque fuselage structure.
(b) Describe the primary advantage of bonded joints over riveted joints.
(c) Identify two critical factors that affect the strength of a riveted joint.

12. Home Assignment and Follow Up	Prepare a brief note comparing the structural assumptions used in engineering theory for the next lesson (LP 10), focusing on load distribution in fuselage structures.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to aircraft structural principles, general assumptions in engineering theory. methods by which structure carries load, typical ideal bending movement and shear force diagram.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min